Exercise 7 On derivatives of exponential and logarithmic functions

Interactive test

From the school book | Understand | Apply | Higher Order Thinking Skills

1 Find the first derivative for each of the following :

(1) $y = e^{2x+3}$

(2) 📖 $y = e^{3x^5}$

(3) 📖 $y = e^{x^2-x}$

(4) $y = e^{(x+1)^2}$

(5) 📖 $y = x^2 e^{-3}$

(6) $y = x^2 + 3e^x$

(7) 📖 $y = 2e^x + \cos 2x$

(8) 📖 $y = 2x + e^{6x}$

(9) $y = x^3 + \tan x + e^{5x+1}$

(10) $y = e^5 + e^{5x} + e^{5x^2}$

2 Find $\frac{dy}{dx}$ for each of the following :

(1) $y = 3^x$

(2) $y = \left(\frac{1}{2}\right)^x$

(3) 📖 $y = 5 \times 6^x$

(4) 📖 $y = 5^{x^3+2x}$

(5) $y = 7^x + x^7$

(6) 📖 $y = 3^{(3x^2-5x+2)}$

(7) 📖 $y = 3e^{\sec x}$

(8) $y = e^{\cos x}$

(9) $y = e^{\sqrt{x}}$

(10) $y = (3^{x-1})^{-2}$

(11) $y = e^{\sin\sqrt{x}}$

(12) $y = e^{\sqrt{\cot x}}$

3 Find the first derivative for each of the following :

(1) $y = x e^{2x}$

(2) 📖 $y = x^2 e^x$

(3) $y = (x^2+3) e^{-3x}$

(4) $y = (3x-5) e^{x^2}$

(5) 📖 $y = e^x \sin x$

(6) $y = e^{2x} \sec 3x$

(7) $y = x \times 5^x$

(8) 📖 $y = 2^x (x+1)$

(9) $y = \sqrt{x}\, e^x$

(10) 📖 $y = e^{\sin x} \times 2^{-5x}$

(11) 📖 $y = e^{2x} a^{x^2-5}$

(12) $y = e^x \times \sin^2 x$

4 Find the first derivative for each of the following :

(1) $y = \frac{x}{e^{2x}}$

(2) 📖 $y = \frac{2e^{x}}{x+1}$

(3) $y = \frac{2^{x}}{x}$

(4) $y = \frac{e^{2x+3}}{x^2+3}$

(5) 📖 $y = \frac{e^{x}}{\tan x}$

(6) $y = \frac{\sin 3x}{e^{x}}$

(7) $y = \frac{e^{\sin x}}{\cos x}$

(8) $y = \frac{e^{-2x}}{\sqrt{x+1}}$

(9) 📖 $y = \frac{e^{x}}{e^{x}+1}$

(10) $y = \frac{e^{x}}{1+\cos x}$

(11) 📖 $y = (e^{3x} - e^{-2x})^5$

(12) 📖 $y = (e^{2x} + e^{-2x})^3$

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5 Find $\frac{dy}{dx}$ for each of the following :

(1) $y = 5 \ln x$

(2) 📖 $y = \ln (2x - 7)$

(3) 📖 $y = 3x + \ln x$

(4) 📖 $y = 5 - 3 \ln x$

(5) 📖 $y = \ln \left(\frac{1}{2}x^2 + x\right)$

(6) 📖 $y = \ln \left(\frac{1}{x^2+9}\right)$

(7) 📖 $y = \log_3 x$

(8) 📖 $y = \log_5 (3x - 2)$

(9) 📖 $y = 2e^{3x} - 5 \log_2 \frac{x}{5}$

(10) 📖 $y = \ln x^2 - e^{\frac{1}{4}x}$

(11) 📖 $y = \log (2x - 3)^2$

(12) $y = 3^{x} + 3 \ln x - 2021$

(13) 📖 $y = \ln \frac{x^2}{x+7}$

(14) 📖 $y = \log \left(\frac{e^{x^2}}{x^2}\right)$

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6 Find $\frac{dy}{dx}$ for each of the following :

(1) $y = x \ln x$

(2) 📖 $y = x^2 \ln x$

(3) 📖 $y = x \ln x^2$

(4) 📖 $y = (2x^5 - 3) \ln x$

(5) 📖 $y = 2x^2 \ln x^3$

(6) $y = 2x + e^{x} \ln x$

(7) 📖 $y = e^{x} \ln (x^2 + 1)$

(8) $y = 5 + 4 \cos x \ln x$

(9) $y = (\log x)^3 + \sqrt{x} \log x^4$

(10) 📖 $y = \frac{x}{\ln x}$

(11) $y = \frac{\ln(5x)}{x+1}$

(12) $y = \frac{\ln x}{\tan x}$

(13) $y = \frac{e^{3x}}{\log x}$

(14) $y = \frac{\log x}{1+\log x}$

(15) $y = \frac{\ln x - 1}{\ln x + 1}$

(16) $y = \frac{1-2\ln x}{\ln x}$

7 Find $\frac{d^2y}{dx^2}$ for each of the following :

(1) $x = e^{2t}$, $y = t^3$ (2) $x = e^{2t}$, $y = e^{3t}$ (3) $x = 6\ln t$, $y = t^2$

8 Find $\frac{d^3y}{dx^3}$ for each of the following :

(1) $y = \ln x$ (2) $y = x^3 + e^{2x}$ (3) $y = x\log x$

9 Find the slope of the tangent to each of the following curves at the indicated points :

(1) $y = \sqrt{x} - 2e^x$, $x = \frac{1}{4}$

(2) $y = x^2 - 3\ln x$, $x = 2$

(3) $y = \frac{1}{4}e^{2x} - 2\ln x$, $x = \frac{1}{2}$

(4) $y = xe^{-x}$, $\left(1 , \frac{1}{e}\right)$

(5) $y = (2x+3)e^{2x}$, $(0 , 3)$

(6) $y = \log_2 5x$, $x = 2$

(7) $y = 4\log(3x+1)$, $x = 1$

(8) $y = \log_2 (2x^2 - 3)^4$, $x = 1$

(9) $y = 3(\log x)^2$, $x = 3$

10 Find the value(s) of x at which the tangent to the following curves is parallel to x-axis , $x > 0$:

(1) $y = x^3 \ln x$

(2) $y = \frac{1}{x^2}\ln x^2$

(3) $y = x^3 - 81\ln x$

(4) $y = 3 - \frac{x^2}{4} + \ln\frac{x}{2}$

11 Find the equation of the tangent to the curve : $y = xe^{2x}$ at the point $\left(\frac{1}{2} , \frac{1}{2}e\right)$

12 Find the equation of the tangent to the curve :

$y = e^{2x}\cos x$ at $x = 0$

13 Find the equation of the tangent to the curve :

$y = 2^x + 2^{-x}$ at the point $\left(2 , 4\frac{1}{4}\right)$

14 **Find the equation of the tangent to the curve :**

$y = \frac{e^{\frac{x}{3}}}{x}$ at the point $\left(3 , \frac{1}{3} e\right)$

15 *(1st session 2017)* **Find the equation of the normal to the curve :**

$y = 3 e^{x}$ at a point lying on it and its x-coordinate equals -1

16 **Find the two equations of the tangent and the normal of the curve :**

$y = x^3 - 18 \ln x$ at a point lies on it and its x-coordinate equals 2

17 *(2nd session 2019)* **Find the equation of the tangent to the curve :**

$y = \ln\left(2 - \sqrt{2} \cos x\right)$ at the point which lying on it and its x-coordinate is $\frac{\pi}{4}$

18 **Find the equation of the normal to the curve :**

$x = e^{t}$, $y = e^{t} + e^{2t}$ at $x = 1$

19 **Choose the correct answer from the given ones :**

(1) If $f(x) = e^{3x}$, then $\grave{f}(x)$ equals .....

(a) e^{2x} (b) $3 e^{3x}$ (c) $9 e^{3x}$ (d) $3 e^{2x}$

(2) *(1st session 2017)* If $f(x) = a e^{x}$, then $\grave{f}(-2)$ equals .....

(a) $-f(2)$ (b) $-\grave{f}(2)$ (c) $-f(-2)$ (d) $f(-2)$

(3) $f(x) = x^2 - 3 \ln 5x$, then $\grave{f}(2)$ equals .....

(a) -1 (b) 1 (c) $\frac{5}{2}$ (d) 6

(4) The equation of the tangent to the curve of the function f where $f(x) = e^{2x+1}$ at the point $\left(\frac{-1}{2} , 1\right)$ is .....

(a) $4y = 3 - 2x$ (b) $y = 2x + 2$ (c) $y = 4x + 3$ (d) $y = 6x + 4$

(5) If $y = \frac{1}{x} \ln e^{x}$, then $\frac{dy}{dx} =$ .....

(a) 1 (b) e^{x} (c) zero (d) $\ln x$

(6) If $y = x^2 \ln e^{x}$, then $\frac{dy}{dx} =$ .....

(a) $3x^2$ (b) x^3 (c) $\ln x^3$ (d) zero

(7) If $y = e^{x}$, $z = \sin x$, then $\frac{dy}{dz} =$ .....

(a) $\frac{e^{x}}{\sin x}$ (b) $e^{x} \tan x$ (c) $e^{x} \cos x$ (d) $\frac{e^{x}}{\cos x}$

(8) *(1st session 2018)* If $a^{y} = b^{x}$ where $a , b \in \mathbb{R}^{+}$, then $\frac{dy}{dx}$ = ...

(a) $\log \frac{a}{b}$ (b) $\log_a b$ (c) $\log_b a$ (d) $\log \frac{b}{a}$

(9) *(1st session 2019)* If $y = (e^{-x} \ln x)$, then $\frac{dy}{dx}$ = ...

(a) $e^{-x}\left(\frac{1}{x} - \ln x\right)$ (b) $e^{x}\left(\frac{1}{x} - \ln x\right)$

(c) $\frac{e^{-x}}{x} - \ln x$ (d) $e^{-x}\left(\frac{1}{x} + \ln x\right)$

(10) The equation of the tangent to the curve $y = b\, e^{-\frac{x}{a}}$ at the point of intersection with the y-axis is .....

(a) $\frac{x}{a} - \frac{y}{b} = 1$ (b) $a x + b y = 1$ (c) $a x - b y = 1$ (d) $\frac{x}{a} + \frac{y}{b} = 1$

(11) If $f(x) = e^{x} g(x)$, $g(\text{zero}) = 2$, $\grave{g}(\text{zero}) = 1$, then $\grave{f}(\text{zero})$ = ...

(a) 1 (b) 3 (c) 2 (d) zero

(12) *(1st session 2022)* If $e^{y} - e\, x = 0$, then $\frac{dy}{dx}$ = ...

(a) $\frac{e}{x}$ (b) $\frac{1}{x}$ (c) $\frac{1}{y}$ (d) $\frac{e}{y}$

(13) *(2nd session 2022)* If a function $f : f(x) = \ln x - (\ln x)^2$ where $x \in]0 , \infty[$, then $\acute{\acute{f}}(1)$ = ...

(a) zero (b) 3 (c) – 3 (d) – 5

(14) The perpendicular to the curve $y = x \ln x$ is parallel to the straight line $2x - 2y + 3 = 0$, then the perpendicular equation is .....

(a) $x - y = 3\, e^{-2}$ (b) $x - y = 6\, e^{-2}$

(c) $x - y = 3\, e^{2}$ (d) $x - y = 6\, e^{2}$

(15) The tangent to the curve $x = e^{\theta} \cos\theta$, $y = e^{\theta} \sin\theta$ at the point where $\theta = \frac{\pi}{4}$ makes with the positive direction of x-axis an angle of measure .....

(a) zero (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

(16) If $\theta \in]0 , \frac{\pi}{2}[$, then the slope of the tangent to the curve $y = \ln(\tan\theta) + \ln(\cos\theta)$, $x = \ln(\sin\theta) + \ln(\cot\theta)$ at $\theta = \frac{\pi}{4}$ equals .....

(a) – 1 (b) zero (c) 1 (d) 2

(17) The measure of the angle that the tangent to the curve $y = e^{\tan x} \sin x$ makes with positive direction of x-axis at $x = \pi$ equals .....

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{2}$ (d) $\frac{3\pi}{4}$

(18) If $f(x) = \grave{f}(x) = e^{g(x)}$ then $g(x)$ could be .....

(a) $x + 2$ (b) x^2 (c) $\ln x$ (d) $2x$

(19) *(Trial 2021)* If $f(x) = \ln(x^2 + 1)^2 + e^{\sin x}$, then $f(0) \times \grave{f}(0) =$ .....

(a) zero (b) 1 (c) $1 + e$ (d) e

(20) *(Trial 2021)* If $y \times \log_{x^3} e = 1$ where $x > 1$, then $\frac{dy}{dx} =$ ..... at $x = 3$

(a) zero (b) 1 (c) $3 \ln 3$ (d) $\ln 3$

(21) *(Trial 2021)* If $y = a\, e^{bx}$, $\frac{d^2y}{dx^2} = y$, then $b^2 =$ .....

(a) zero (b) 1 (c) -1 (d) 2

(22) *(Trial 2021)* If equation of tangent to the curve $y = -\ln x$ at a point lies on it (a , b) is $y = m x$, then $a =$ .....

(a) e (b) $\frac{1}{2}$ e (c) $\frac{3}{2}$ e (d) $2\sqrt{e}$

(23) If $x = e^{\sin\theta}$, $\frac{dy}{d\theta} = 2^{\log_2 \cos\theta}$, then $\frac{dy}{dx} =$ ..... at $\theta = 0$

(a) $\frac{2}{e}$ (b) 1 (c) e (d) $\frac{1}{e}$

(24) *(2nd session 2021)* If $x^2 = e^{2y}$, then $x \times \frac{dy}{dx} =$ .....

(a) -1 (b) 1 (c) x (d) e^{2x}

(25) *(Trial 2021)* Slope of tangent to the curve $y = 5^x \log_5 (x + 1)$ at $x = 0$ equals .....

(a) $\ln 5$ (b) $\log_5 e$ (c) zero (d) $5 \log_5 e$

(26) *(2nd session 2021)* If the equation of the tangent to the curve : $\log_y (4x) = 2x$ at the point $(1 , 2)$ which lies on it is $y = mx + c$, then $m =$ .....

(a) $1 - 2 \ln 2$ (b) $2 \ln 2 - 1$ (c) $2 \ln 2 + 1$ (d) $2 + \ln 2$

(27) *(2nd session 2021)* The slope of the normal to the curve $x = e^t$, $y = e^{2t+2}$ at the point which lies on the curve and its x-coordinate equals one is .....

(a) $-2 e^2$ (b) $\frac{-1}{2e^2}$ (c) $2 e^2$ (d) $\frac{e^2}{2}$

(28) *(2nd session 2022)* The equation of the normal to the curve $e^y = x^2$ at the point which lies on the curve and its x-coordinate = 1 is .....

(a) $2x = 2 - y$ (b) $y = 2x - 2$ (c) $x = 2y + 1$ (d) $2y = 1 - x$

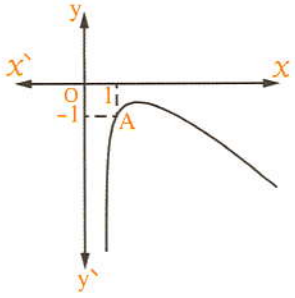
(29) *(1st session 2022)* If the two curves $y_1 = ae^{-x}$, $y_2 = be^{x}$ are intersecting orthogonally, then .....

(a) $ab = -1$ (b) $a = 2b - 1$ (c) $a = b - 1$ (d) $ab = 1$

(30) If $f(x) = \ln\left(x \,.\, g(x)\right)$ and $g(4) = \grave{g}(4) = 4$, then $\grave{f}(4) =$ .....

(a) 1 (b) $\frac{5}{4}$ (c) $\frac{3}{2}$ (d) 2

(31) *(Trial 2021)* The opposite figure represents the curve $e^{x+y} = 3x - 2$ if the tangent to the curve at the point A $(1, -1)$ intersects the two coordinate axes $\overleftrightarrow{xx}$, $\overleftrightarrow{yy}$ at the two points B and C respectively , then A divides $\overrightarrow{BC}$ by the ratio ..... internally.



(a) 1 : 2 (b) 2 : 1 (c) 3 : 2 (d) 2 : 3

20 If $e^x = \sum_{n=0}^{\infty} \frac{x^n}{\lfloor n}$ **Prove that :** $\frac{d}{dx}(e^x) = e^x$

21 If $y = x^2 \ln \frac{x}{a}$ **Find :** $\frac{d^3y}{dx^3}$ at $x = 4$ « $\frac{1}{2}$ »

22 If $y = \ln(1 + e^x)$, find value of x which makes : $e^y \frac{dy}{dx} = 6$ « ln 6 »

23 If $x = y^2 e^{\sqrt{y}}$ **Find :** $\frac{dx}{dy}$

24 If $x = y^2 \ln y$, $y > 0$ **Find :** $\frac{dy}{dx}$

25 If $y = \ln(\csc x - \cot x)$ **Prove that :** $\frac{dy}{dx} = \csc x$

26 If $y = \ln\left(\tan\left(\frac{\pi}{4} + \frac{x}{2}\right)\right)$ **Prove that :** $\frac{dy}{dx} = \sec x$

27 If $y = \ln\left(\sqrt{\sec x}\right)$ **Prove that :** $\frac{dy}{dx} = \frac{1}{2} \tan x$

28 If $y = \ln\left(x + \sqrt{x^2 + a^2}\right)$ **Prove that :** $\frac{dy}{dx} = \frac{1}{\sqrt{x^2 + a^2}}$

29 If $y = e^{2x} \sin 2x$ **Prove that :** $\frac{d^2y}{dx^2} = 8y \cot 2x$

30 If $y = e^{3x} + x^2$ **Prove that :** $\frac{d^2y}{dx^2} - 2 = 9(y - x^2)$

31 If $y = a\, e^{x^2+1}$ **Prove that :** $\frac{d^3y}{dx^3} = 4xy(3 + 2x^2)$

32 If $y = a\, e^{\frac{b}{x}}$ **Prove that :** $x y \dddot{y} + 2y \ddot{y} - x \dot{y}^2 = 0$

33 If $y = \sqrt{\frac{x^2+1}{x^2-1}}$ **Prove that :** $(x^4-1)\,\dot{y} + 2x\,y = 0$

34 If $y = e^{-x}\sqrt{\frac{1+x}{1-x}}$ **Prove that :** $(1-x^2)\,\dot{y} = x^2\,y$

35 Find rate of change of $\ln(9+x^3)$ with respect to (x^2+3) at $x=1$ « $\frac{3}{20}$ »

36 If $y = e^{ax}$

(1) **Prove that :** $\frac{d^4y}{dx^4} - a^4\,y = 0$

(2) Find the values of « a » which satisfied $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = 0$ « 1 »

37 *(2nd session 2018)* If $x\,e^y = 2 - \ln 2 + \ln x$, $\frac{dx}{dt} = 6$ at $x=2$, $y =$ zero , **find :** $\frac{dy}{dt}$ « $-\frac{3}{2}$ »

38 **Find $\frac{dy}{dx}$ for each of the following :**

(1) $y = \sec e^x$

(2) $y\,e^x = e^3$

(3) $x \ln y = 58$

(4) $y = \ln|x^3 - 7x^2 - 3|$

(5) $y = 2^{\sec^2 x}$

(6) $y = \log|\tan x|$

(7) $y = \sin(\log x)$

(8) $y = e^{\sin x} \times \sin e^x$

(9) $y = \frac{1}{e^x \tan x}$

(10) $y = \sqrt{\frac{1+e^x}{1-e^x}}$

(11) $y = \ln(\ln x)$

(12) $y = \ln\big(\ln(\ln x)\big)$

(13) $y = \ln(1+e^{3x})$

(14) $y = \log\sqrt{x}$

(15) $y = \sqrt{\log x}$

(16) $y = [x + \ln(2x)]^3$

(17) $y = \ln[\sin(e^x + 5x + 8)]$

(18) $y = \log\sqrt{\frac{x-1}{x+1}}$

(19) $y = \frac{e^x \ln x}{x^2}$

(20) $y = \log(\cos x^2)$

(21) $y = \cos(\log x)^2$

(22) $y^2 = e^{5x^2-3}$

(23) $y^2 = 3^x \times 2^y$

(24) $y = 4^{3\sin x - e^x}$

(25) $y = \pi^{\sin x} + e^{\pi}$

(26) $y = e^{3x+2\ln x}$

(27) $y = \ln(\sec x + \tan x)$

(28) $y = \ln\left(\frac{\ln x}{x}\right)$

39 **Find $\frac{dy}{dx}$ for each of the following :**

(1) $y = x^{2x}$

(2) *(2nd session 2017)* $y = (x^3 + 5)^x$

(3) $y = 3x^{x^2+2}$

(4) $y = x^{\frac{1}{x}}$

(5) $y = x^{\sin x}$

(6) $y = (\sin x)^x$

(7) $y = (1 - 3x)^{\cos x}$

(8) $y = (\sin x)^{\tan x}$

(9) $y = x^{e^x}$

(10) $y = e^{x^e}$

(11) $y = e^{e^x}$

(12) $y = (\log x)^{\log x}$

(13) $y = (\log x)^{\tan x}$

(14) $y = (x^2 + 3)^{\log x}$

(15) $y = (\log x)^x$

(16) $y = x^{x^x}$

40 If $\overleftrightarrow{AB}$ is a tangent to the curve $y = \ln \frac{x}{2}$ at point C $(1, y)$ and intersects x-axis at point A and y-axis at point B , find the length of $\overline{AB}$ « $\sqrt{2}$ ln 2 e length unit »

41 If the normal to the curve $y = \ln 2x$ at point A $(1, \ln 2)$ intersects x-axis at point B , find the length of $\overline{AB}$ to the nearest three decimals. « 0.980 length unit »

42 If the tangent of the curve $y = e^x$ at point $(2, e^2)$ intersects x-axis at point A and y-axis at point B , find the length of $\overline{AB}$ « 7.46 length unit »

43 If the daily production of a factory during a certain period of time t (day) is identified by the relation $y = 400 (1 - e^{-0.3t})$ units , find the rate of change in the number of the produced units with respect to time on the tenth day. « $120 e^{-3}$ »

44 If the production of a beehive is given by the relation : $y = (t + 100) \ln (t + 5)$ grams in terms of the number of days t , find the rate of change of the beehive production when $t = 5$, $t = 15$, $t = 20$ Does the production of the beehive increase or decrease ?

« 12.8 , 8.7 , 8.02 , decreased »

45 **Choose the correct answer from the given ones :**

(1) The rate of change of e^{x^3} with respect to $\ln x$ equals .....

(a) $3x^3 e^{x^3} + 3x^2$ (b) e^{x^3}

(c) $3x^3 e^{x^3}$ (d) $3x^2 e^{x^3}$

(2) The curve $y - e^{xy} + x = 0$ has a vertical tangent at the point .....

(a) $(1, 1)$ (b) $(0, 1)$ (c) $(1, 0)$ (d) $(2, e^2)$

(3) *(2nd session 2022)* If the function $f : f(x) = \ln(\sin x) - \ln(\cos x)$ where $x \in]0 , \frac{\pi}{2}[$, then $\grave{f}\left(\frac{\pi}{4}\right) =$ .....

(a) 2 (b) zero (c) 1 (d) 3

(4) Let $f(x) = (\cos x)^{\cos x}$, then $\grave{f}$ (zero) = ...

(a) – 3 (b) – 2 (c) – 1 (d) zero

(5) Let $f(x) = e^{\tan x}$, then $\lim\limits_{x \to \frac{\pi}{4}} \dfrac{f(x) - f\left(\frac{\pi}{4}\right)}{x - \frac{\pi}{4}} =$ .....

(a) e (b) 2 e (c) e^2 (d) $2e^2$

(6) Let $y = \ln(\sin x)$, then $\dfrac{d^2 y}{dx^2} =$ .....

(a) $-\csc^2 x$ (b) $\sec x$ (c) $-\csc x \cot x$ (d) $\sec x \tan x$

(7) Let $f(x) = e^{\ln x}$, then $\grave{f}(x) =$ .....

(a) $(\ln x)\, e^{\ln x}$ (b) $e^{\ln x}$ (c) $(\ln x)\, e^{x - \ln x}$ (d) 1

(8) Let $f(x) = e^{\ln(x^3 - 2x + 1)}$, then $\grave{f}(0) =$ .....

(a) – 4 (b) – 2 (c) zero (d) 2

(9) Let $f(x) = x - x \ln x$, then the slope of the tangent to the curve at $x = e$ equals .....

(a) zero (b) – 1 (c) 1 (d) e

(10) If $\sin x = e^y$ where $0 < x < \pi$, then $\dfrac{dy}{dx} =$ .....

(a) $\tan x$ (b) $\cot x$ (c) $-\tan x$ (d) $-\cot x$

(11) *(2nd session 2018)* If $y = e^{(1 + \ln x)}$, then $\dfrac{dy}{dx} =$ .....

(a) x (b) $e\,x$ (c) e (d) 1

(12) The ratio between slope of the tangent of $y = \ln\left(3\sqrt{x+1}\right)$ and slope of the tangent of $y = \ln\left(5\sqrt{x+1}\right)$ at $x = a$ is .....

(a) 3 : 5 (b) 5 : 3 (c) 1 : 1 (d) ln 3 : ln 5

(13) If $y = \ln x$, n is positive integer , then $\dfrac{d^n y}{dx^n} =$ .....

(a) $\left(\dfrac{-e}{x}\right)^n$ (b) $(n-1)\, x^{-n}$

(c) $(n+1)\, x^{-n-1}$ (d) $(-1)^{n-1} \lfloor (n-1)\, x^{-n}$

(14) If $f(x) = x \ln x$ and $\dfrac{d^8 y}{dx^8} = \dfrac{k}{x^7}$, then k = ...

(a) $\lfloor 5$ (b) $\lfloor 6$ (c) $\lfloor 7$ (d) $\lfloor 8$

(15) If $y = \sqrt{e^{ax}}$ and $\ddot{y} + 4\dot{y} + 4y = 0$, then $a =$ .....

(a) 4 (b) – 4 (c) 16 (d) – 16

(16) If $x = a e^{\theta}(\sin\theta - \cos\theta)$, $y = a e^{\theta}(\sin\theta + \cos\theta)$, then $\left(\frac{dy}{dx}\right)_{\theta=\frac{\pi}{4}} =$ .....

(a) 1 (b) 2 (c) $\frac{2}{\pi}$ (d) $\frac{1}{\pi}$

(17) If $y = 4^{\log_2 \sin x} + 9^{\log_3 \cos x}$, then $\frac{dy}{dx} =$ .....

(a) zero (b) 1 (c) $\sin x + \cos x$ (d) – 1

(18) If $k, m \in \mathbb{R}$, $f(x) = x e^{x}$ and $f^{(10)}(x) = k e^{x} + m x e^{x}$, then $k + m =$ .....

(a) 9 (b) 10 (c) 11 (d) 12

(19) If $y = x^{x}$, then $\frac{dy}{dx} =$ .....

(a) $1 + \ln x$ (b) x^{x} (c) $x^{x} \ln x$ (d) $x^{x}(1 + \ln x)$

(20) *(Trial 2021)* The slope of the curve $x^{y} - y^{x} = 0$ at the point $(1, 1)$ which lies on it equals .....

(a) 1 (b) zero (c) 2 (d) – 1

(21) If $y = \ln \frac{e^{x^2}}{x^2}$, then $\frac{dy}{dx} =$ zero at $x =$ .....

(a) zero (b) ± 3 (c) ± 2 (d) ± 1

(22) *(1st session 2021)* The equation of the normal to the curve $y = \ln(\tan x)$ at the point which lies on the curve and its x-coordinate equals $\frac{\pi}{4}$ is .....

(a) $4x - 8y = \pi$ (b) $8y + 4x = \pi$ (c) $4x + 2y = \pi$ (d) $4x - 2y = \pi$

(23) *(1st session 2022)* The equation of the normal to the curve : $x^{y-x} = 1$ at the point which lies on the curve and its x-coordinate equals (e) is .....

(a) $3y - x = 2e$ (b) $3x - y = 2e$ (c) $3y - 2x = e$ (d) $y + x = 2e$

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46 If $e^{x} y = x e^{y}$ **Find :** $\frac{dy}{dx}$ « $\frac{y - xy}{x - xy}$ »

47 If $e^{2x} \ln y = x + y - 2$ **Find :** $\frac{dy}{dx}$ at point $(1, 1)$ « $\frac{1}{e^2 - 1}$ »

48 If $x e^{xy} = y + \sin^2 x$ **Find :** $\frac{dy}{dx}$ at $x = 0$ « 1 »

49 If $e^{xy} - x^2 + y^3 = 0$ **Find :** $\frac{dy}{dx}$ at $x = 0$ « $\frac{1}{3}$ »

50 If $x e^{-\frac{1}{2}y} + y e^{-\frac{x}{2}} = 2$ **Find :** $\frac{dy}{dx}$ at $x = 0$ « $\frac{e - 1}{e}$ »

51 If $x^2 y = a b \ln x$ **Prove that :** $x^2 \ddot{y} + 5x\dot{y} + 4y = 0$

52 If $y \ln(xy) = x$ **Prove that :** $\frac{dy}{dx} = \frac{y(x - y)}{x(x + y)}$

Higher skills

53 **Choose the correct answer from those given :**

(1) If $y = \left(1 + \frac{x}{\lfloor 1} + \frac{x^2}{\lfloor 2} + \frac{x^3}{\lfloor 3} + \cdots\right)$, then $\frac{d^3 y}{dx^3} =$ .....

(a) $\lfloor 3\, x$ (b) $\lfloor 3\, y$ (c) x (d) y

(2) If $f(x) = e^x \sin x$, $g(x) = e^x \cos x$ which of the following statements is not true ?

(a) $\dot{f}(x) = f(x) + g(x)$ (b) $\dot{g}(x) = g(x) - f(x)$

(c) $\ddot{f}(x) = 2\, g(x)$ (d) $\ddot{g}(x) = 2\, f(x)$

(3) If $f : \mathbb{R} - \left\{ \frac{\pi}{2} + n\pi , n \in \mathbb{Z} \right\} \longrightarrow \mathbb{R}$ where $f(x) = e^{\tan x}$, then for all values of x we find .....

(a) $\dot{f}(x) \geq f(x)$ (b) $\dot{f}(x) \leq f(x)$ (c) $\dot{f}(x) > f(x)$ (d) $\dot{f}(x) < f(x)$

(4) If the function g is the inverse function of f where $f(x) = x^3 + 3x + e^x$, then $\dot{g}(1) =$ .....

(a) zero (b) 1 (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

54 If $x = e^{\cos 2t}$, $y = e^{\sin 2t}$ **Prove that :** $\frac{dy}{dx} = \frac{-y \ln x}{x \ln y}$

55 If $y = a \cos(\ln x) + b \sin(\ln x)$ **Prove that :** $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = 0$

56 If $y = x^x$ **Prove that :** $\ddot{y} = x^x (1 + \ln x)^2 + x^{x-1}$

57 If $y = \frac{1}{2} \ln\left(\frac{1 - \cos 2x}{1 + \cos 2x}\right)$ **Prove that :** $\frac{dy}{dx} = 2 \csc 2x$

58 If $y = \ln \frac{\sqrt{x^2 + 1} - 1}{\sqrt{x^2 + 1} + 1}$ **Prove that :** $\frac{dy}{dx} = \frac{2}{x\sqrt{x^2 + 1}}$

59 If $y = x^{x^{x \cdots}}$ **Prove that :** $x \frac{dy}{dx} = \frac{y^2}{1 - y \ln x}$

60 If $y = \sqrt{\ln x + \sqrt{\ln x + \sqrt{\ln x + \cdots}}}$

Prove that : $(2y - 1) \frac{dy}{dx} = \frac{1}{x}$